

Maturing Through Crises

A Macroeconomic Analysis of Indonesia in Perspective to ASEAN (1995–2024) Through the Lenses of SDG 8: Decent Work and Economic Growth

1. Project Details

Dataset: IMF World Economic Outlook, April 2024

Period: 1995 – 2024

Coverage: 9 ASEAN countries, excluded Myanmar and Timor-Leste

2. Key Findings

The 1998 Asian Financial Crisis caused Indonesia to suffer the worst GDP contraction amongst other ASEAN countries (-13.1% YoY) along with a terrible hyperinflation (approximately 58% YoY), due to the significant depreciation of the Indonesian Rupiahs, plunging from Rp2,441 per dollar down to Rp16,800 per dollar (-85.83%). The crisis then led to the creation of statutory 3% budget deficit ceiling through UU No. 17/2003. However, in the 2008 Global Financial Crisis, Indonesia was one of the few countries that managed to maintain positive GDP growth (~4.7% YoY) and successfully maintained inflation levels at ~4.8 (YoY), benefitting from a large domestic consumer-base and a disciplined monetary and fiscal policies and proving that we learned from our costly flaws in 1998.

Indonesia successfully maintained a relatively low Gross Debt-to-GDP ratio across the last 20 years, especially compared to Singapore's consistently high Debt-to-GDP ratio. However, it is also important to note that Singapore's high Debt-to-GDP ratio is used for investment (via the Temasek and GIC) and reserves management, unlike Indonesia's consistent debt use to finance budget deficits. This budget deficit is because Indonesia's government revenue only makes up of 14.1% of GDP. This terrible government revenue-to-GDP percentages is the lowest in ASEAN, 420 basis points lower than ASEAN's average (excluding Timor-Leste), reflecting the massive challenge of revenue-gathering Indonesia is facing, either from tax or from state-owned companies (BUMN).

Although investment alone only explain a small part of the growth variance, Indonesia's investment efficiency is still terrible. Across 2019-2024, Indonesia's average investment composition of the GDP is one of the highest in ASEAN (30.1%, 270 basis points above the 27.4% ASEAN average), but its average GDP growth rate is one of the lowest (4.4%, 10 basis points lower than the 4.5% ASEAN average). This suggests that Indonesia's facing a challenge in converting capital into productive quality output, which is a key SDG 8 concern.

Despite all the bad news, Indonesia's GDP is the largest in ASEAN (1.37Trillion USD), yet our PPP per capita is lower than ASEAN's average, caused by our massive population. Though this is most likely to happen to developing countries and emerging markets, it is still important to note because it involves the welfare of the middle-class.

3. Methodology Overview

3.1. Data Source: IMF World Economic Outlook (WEO) Database, April 2024 Edition, covering 9 ASEAN economies (excluded Myanmar and Timor-Leste)

3.2. Scope: Annual data, 1995-2024, covering GDP growth, inflation, GDP per capita (in USD & Purchasing Power Parity-adjusted value), government debt, fiscal balance, government revenue/expenditure, and total investment.

3.3. Preparation: NULL detection/audit by year, country, and indicator. IMF 2025-2027 is entirely dropped to avoid using biased projections.

3.4. Analysis Framework: Analysis is done in respect to 3 major financial crises: 1998 Asian Financial Crisis, 2009 Global Financial Crisis, and 2020 COVID-19 Pandemic.

3.5. Visualization: 10 main charts for macroeconomic insights (multi-line time-series, dual-axis overlay, double bar plot, heatmap, scatter + OLS regression, horizontal bar, bubble chart), and 3 extra charts for NULL detection (bar plots). All visualizations done using python 3.13.5 using matplotlib, scipy, and seaborn.

4. SDG Target Mapping

4.1. SDG 8.1: Sustain per-capita growth, achieved through GDP-per-capita trend visualization (figure 3.1 & 7.1)

4.2. SDG 8.2: Higher productivity/investment, achieved through investment-growth scatter plot and investment efficiency analysis (figure 6.1 & 6.2)

4.3. SDG 8.10: Strengthen financial institutions, achieved through fiscal balance (revenue-expenditure) analysis (figure 5.1 & 5.3)

5. Conclusions

Indonesia's macroeconomic conditions throughout 1995-2024 tells a story of learning, from 1998's catastrophe, 2008's miracle, 2020's black swan, into today's stable 5% GDP growth with managed inflation. Crisis response has been immediate and stable, proven by GDP growth resilience & manageable inflation shown across crises. However, the investment efficiency gap (Indonesia ranks 6th out of 9 in GDP growth per unit investment, despite above average capital use of 30.1% of GDP) signals the need for structural fixes for efficiency rather than capital. The low government revenue composition challenge (tax-to-GDP ~14-16%, 590 basis points below ASEAN's average) limits GDP growth and public investment in human resources & infrastructure, which is a challenge addressed in SDG 8.2.

6. Limitations

6.1. Country Exclusion: Excluded Myanmar, Timor Leste in some analyses due to data sparsity.

6.2. Year Exclusion: excluded IMF's 2025-2027 projected data from all charts to prevent bias.

6.3. Exchange Rate: Nominal USD figures fluctuate, but PPP adjusts this partially.

7. Data Source

7.1. International Monetary Fund. (2024). World Economic Outlook Database, April 2024. Washington, D.C.: IMF. Retrieved from <https://www.imf.org/en/Publications/WEO/weo-database/2024/April>

7.2. United Nations. (2015). Transforming Our World: The 2030 Agenda for Sustainable Development (Resolution A/RES/70/1). UN General Assembly. — SDG 8: Decent Work and Economic Growth.

7.3. Republic of Indonesia. (2003). Undang-Undang No. 17 Tahun 2003 tentang Keuangan Negara [Law No. 17/2003 on State Finance — 3% deficit ceiling]. State Secretariat, Jakarta.

7.4. Republic of Indonesia. (2020). Undang-Undang No. 2 Tahun 2020 tentang Kebijakan Keuangan Negara dan Stabilitas Keuangan Negara.